

HARBEL, Liberia

WMO: 656600

Lat: 6.234N

Lon: 10.362W

Elev: 31

StdP: 14.68

Time Zone: 0.00 (E00)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	69.8	71.4	64.3	90.3	76.3	67.7	101.8	77.1	17.1	79.9	16.3	80.2	3.0	40	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	15.9	91.8	80.3	91.5	80.3	90.0	79.9	82.7	89.6	81.9	88.7	81.3	87.9	9.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.8	160.3	87.6	80.3	157.5	87.0	79.2	151.7	85.5	46.5	89.7	45.7	89.5	45.1	88.6	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
15.7	14.1	12.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	81.5	82.6	82.8	82.4	81.6	79.5	78.1	77.6	78.7	80.3	81.3	81.5
	DBStd	2.35	1.92	1.73	1.45	1.53	1.62	1.82	1.53	1.47	1.60	1.59	1.27	1.50
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11187	978	914	1017	972	980	885	870	855	861	940	939	976
	CDD65	5712	513	494	552	522	515	435	405	390	411	475	489	511
	CDH74	48060	4490	4649	5495	5211	4882	3342	2495	2159	2684	3726	4375	4552
	CDH80	15208	1822	1854	2237	1984	1623	687	307	210	442	974	1402	1666
(14) Wind	WSAvg	4.2	3.7	4.1	4.4	4.4	4.3	4.3	4.6	4.8	4.3	4.2	3.7	3.4
(15) Precipitation	PrecAvg	90.3	1.1	2.0	2.8	5.0	8.9	14.5	10.0	11.5	14.2	11.8	6.6	2.0
	PrecMax	115.9	5.9	3.7	4.4	8.7	13.2	21.0	20.2	18.6	21.9	17.7	10.8	4.3
	PrecMin	60.2	0.2	0.9	1.4	2.0	4.2	7.5	5.3	3.9	7.9	6.0	3.7	0.4
	PrecStd	11.0	1.1	0.7	0.7	1.7	2.4	3.2	3.7	3.5	3.5	2.8	1.8	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.2	93.6	93.5	91.8	91.3	88.1	86.0	85.7	87.6	89.7	91.0	91.7
		MCWB	77.7	80.0	80.7	81.8	81.8	80.4	78.8	78.6	80.5	80.6	81.4	79.7
	2%	DB	91.5	91.8	91.8	91.3	89.8	86.3	84.3	83.9	85.8	88.0	89.4	90.0
		MCWB	78.9	80.0	80.7	81.5	80.8	79.5	77.6	77.4	79.0	80.3	80.6	79.3
	5%	DB	89.9	91.2	91.2	89.8	89.1	85.7	82.7	82.4	84.2	86.4	88.0	89.5
		MCWB	78.7	79.8	80.6	80.7	80.6	79.2	77.0	76.9	78.2	79.3	80.3	79.2
	10%	DB	89.2	89.6	89.7	89.3	87.7	84.2	82.1	80.8	82.6	85.6	86.4	87.9
		MCWB	78.5	79.0	79.8	80.5	80.4	78.7	76.8	76.5	77.9	79.0	79.4	79.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.6	82.9	83.3	84.0	82.7	81.0	79.8	79.6	80.6	81.9	82.3	82.6
		MCDB	90.1	91.4	91.0	90.8	88.4	85.5	83.6	83.1	85.2	87.7	89.0	89.2
	2%	WB	81.1	81.7	82.2	82.5	81.7	80.3	78.9	78.5	79.6	80.6	81.0	81.4
		MCDB	88.5	89.8	89.8	89.0	87.6	84.9	82.9	82.1	84.0	86.3	87.4	88.0
	5%	WB	80.1	80.6	81.3	81.6	80.9	79.7	78.1	77.8	79.0	79.9	80.3	80.5
		MCDB	87.2	88.1	88.6	88.1	86.7	84.2	82.0	81.2	83.1	85.3	86.4	87.0
	10%	WB	79.5	79.7	80.3	80.7	80.2	79.0	77.3	77.1	78.3	79.3	79.8	79.9
		MCDB	86.3	86.8	87.2	87.2	85.8	83.3	80.9	80.3	82.1	84.3	85.6	85.9
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	15.9	15.9	14.6	12.8	9.9	8.1	7.4	9.2	11.7	13.0	15.0
		MCDBR	18.0	17.4	16.9	15.5	14.2	11.8	9.9	9.5	11.2	13.2	14.1	16.4
	5% WB	MCWBR	8.1	7.3	7.6	7.5	7.0	6.2	5.4	5.3	6.1	7.0	7.1	7.6
		MCDWR	16.4	16.1	15.8	15.0	13.3	10.8	9.1	8.5	10.2	12.7	13.6	14.8
(40) Clear-Sky Solar Irradiance	taub	taub	0.738	0.723	0.737	0.599	0.521	0.513	0.496	0.481	0.466	0.487	0.551	0.654
		taud	1.618	1.673	1.656	1.946	2.148	2.174	2.230	2.282	2.342	2.269	2.049	1.788
	Ebn at Noon	Ebn at Noon	201	210	208	235	249	247	253	262	270	265	245	217
		Edh at Noon	84	81	83	61	49	47	44	43	41	44	54	70
(44) All-Sky Solar Radiation	RadAvg	1576	1601	1642	1616	1499	1256	1226	1212	1302	1472	1473	1484	
	RadStd	102	91	101	86	64	73	68	77	60	60	66	73	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page